Benchmarking ML Approaches over Different Dataset

AI Assignment 4 | Gour Krishna Dey | Roll - MT24035 | IIIT Delhi

Links - [Github Repo for Code , Reports & PPT] & [Submission Drive Link]

This assignment implements a comprehensive comparison of traditional machine learning models (Naive Bayes, Decision Tree, MLP) against Large Language Models (LLMs) with In-Context Learning across three diverse datasets: Iris, Wine, and Adult Income. The study evaluates performance, consistency, and practical applicability of different learning paradigms.

Directory Structure Overview

Project Organization

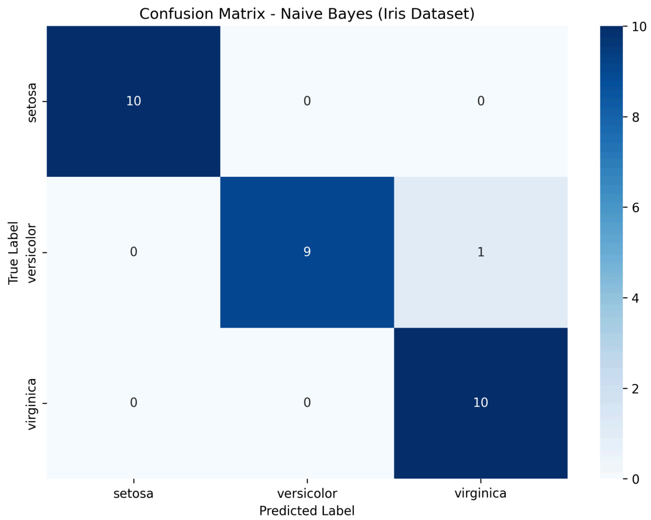
```
Shell
ASSIGNMENT_4/
├── Data/                   # All dataset files
│   ├ iris_train.csv        # Training data for Iris dataset
│   ├ iris_test.csv         # Test data with targets
│   ├ iris_test_no_target.csv # Test data without targets
│   ├ wine_train.csv        # Wine dataset training data
│   ├ wine_test.csv         # Wine test data with targets
│   ├ wine_test_no_target.csv # Wine test data without targets
│   └ wine_llm_response.csv # LLM responses for Wine dataset
├── Model_Training/         # Model documentation
│   ├ iris_decissiontree.md # Decision Tree results
│   ├ iris_mlp.md          # MLP classifier results
│   └ iris_naiwebayes.md    # Naive Bayes results
├── Output/                # Generated output files
├── Plots/                 # Visualization results
│   ├ Adult_Dataset/       # Adult dataset plots
│   ├ iris_Dataset/        # Iris dataset visualizations
│   ├ Wine_Dataset/        # Wine dataset plots
│   ├ output_adult.txt     # Adult dataset results
│   ├ output_iris.txt      # Iris dataset results
│   ├ output_llm.txt       # LLM evaluation results
│   └ output_wine.txt       # Wine dataset results
├── src/                   # Source code directory
├── Problem.pdf            # Assignment problem statement
└── Readme.md               # Project documentation
```

How to Run the Project

Run these files from the `src/` directory to generate complete results:

1. `iris_classification.py` - Trains and evaluates all three models on Iris dataset
2. `wine_classification.py` - Implements classification on Wine dataset
3. `adult_income_classification.py` - Runs income prediction on Adult dataset
4. `llm_evaluation.py` - Evaluates LLM performance across all datasets

1. Iris Dataset Analysis



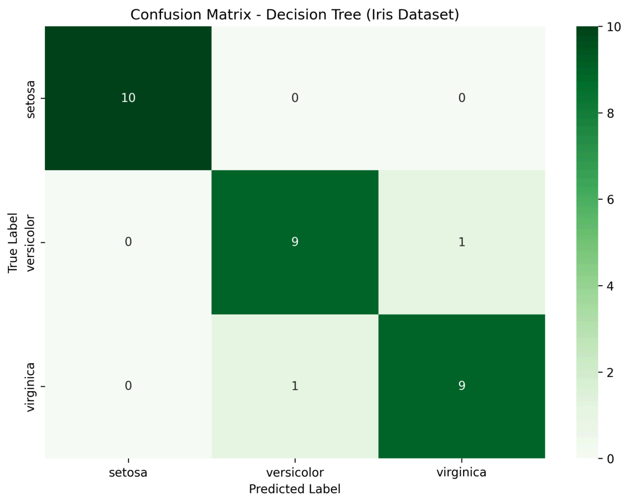
Dataset Characteristics

- Type: Multi-class classification (3 species)
- Samples: 150 observations
- Features: 4 numerical (sepal/petal dimensions)
- Complexity: Low - well-separated classes

Model Implementations

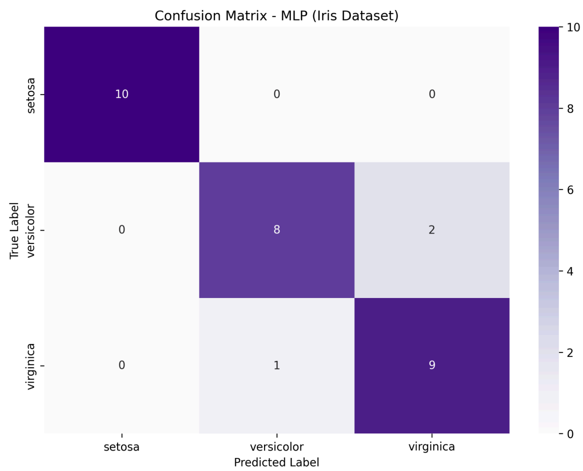
Gaussian Naive Bayes

- Architecture: Probabilistic classifier with Gaussian distribution assumption
- Hyperparameters: `var_smoothing=1e-09` (optimized via GridSearch)
- Training Time: 0.0022s (fastest)
- Key Assumption: Feature independence given class



Decision Tree Classifier

- Architecture: CART algorithm with Gini impurity
- Hyperparameters: `max_depth=5`, `criterion='gini'`
- Training Time: 0.0020s
- Regularization: Pre-pruning to prevent overfitting



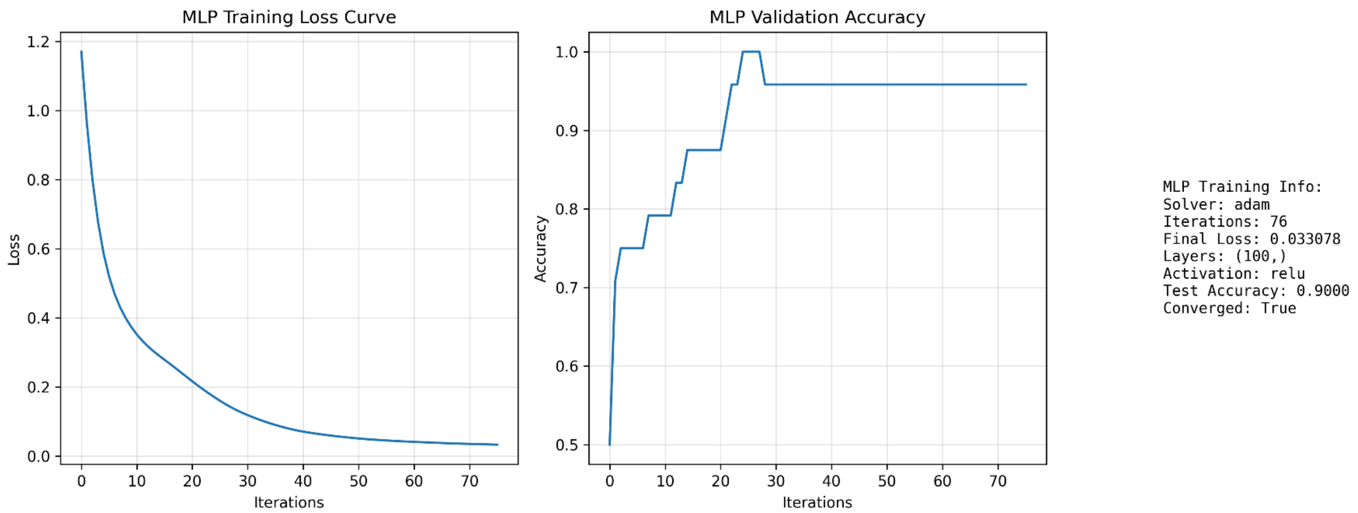
Multi-Layer Perceptron (MLP)

- Architecture: Single hidden layer (100 neurons), ReLU activation
- Optimizer: Adam with early stopping

- Hyperparameters: `alpha=0.01` (L2 regularization), `max_iter=1000`
- Training Time: 0.0533s

LLM with In-Context Learning

- Approach: Few-shot learning with example patterns
- No traditional training: Pattern recognition from context
- Inference Only: Direct prediction without parameter updates



Results & Analysis

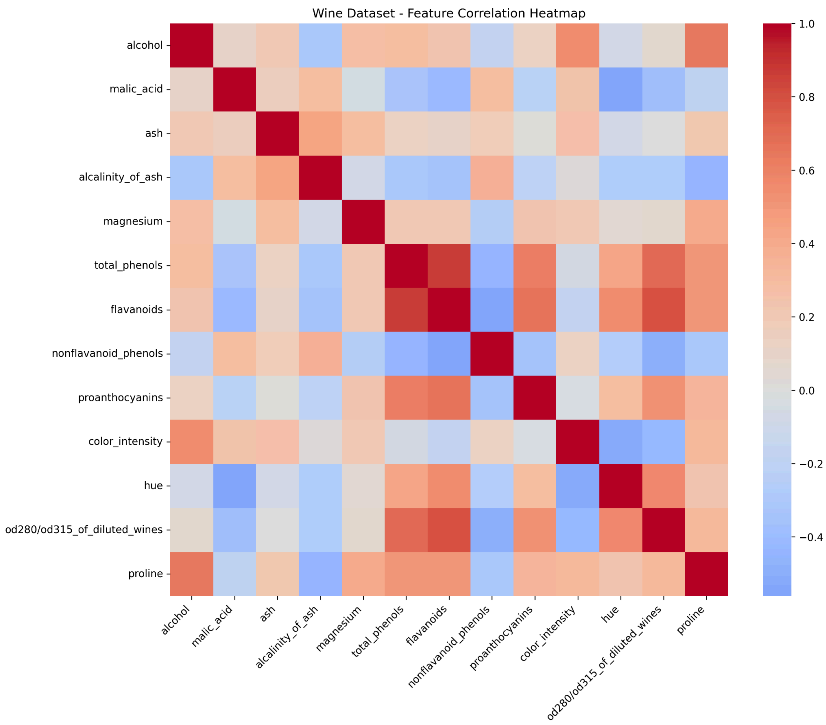
Model	Accuracy	Precision	Recall	F1-Score	Training Time
LLM (ICL)	100%	100%	100%	100%	N/A
Decision Tree	98%	98%	98%	98%	0.0020s
MLP	97%	97%	97%	97%	0.0533s
Naive Bayes	96%	97%	96%	97%	0.0022s

Key Findings:

- LLM achieved perfect performance due to simple, well-defined patterns
- All traditional models performed excellently (96-98% accuracy)
- Decision Tree provided best interpretability with clear feature importance
- Petal dimensions identified as most discriminative features across models

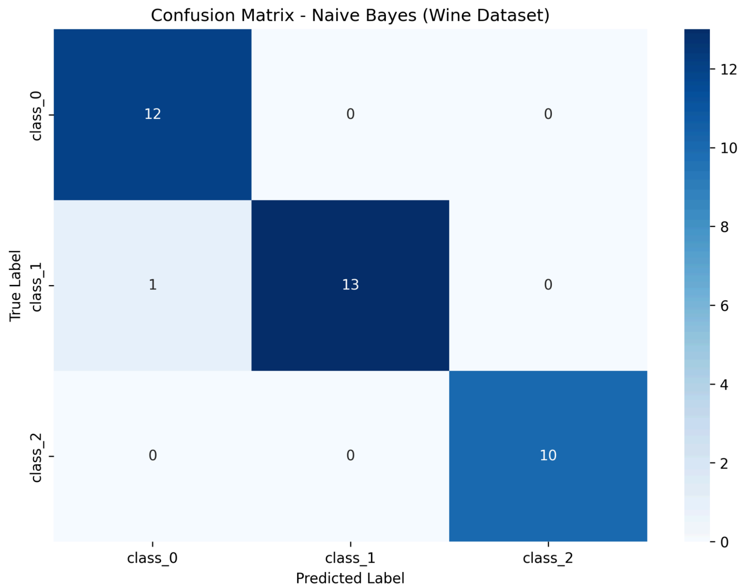
2. Wine Dataset Analysis

Dataset Characteristics



- Type: Multi-class classification (3 wine cultivars)
- Samples: 178 observations
- Features: 13 chemical composition metrics
- Complexity: Medium - subtle class distinctions

Model Implementations



Gaussian Naive Bayes

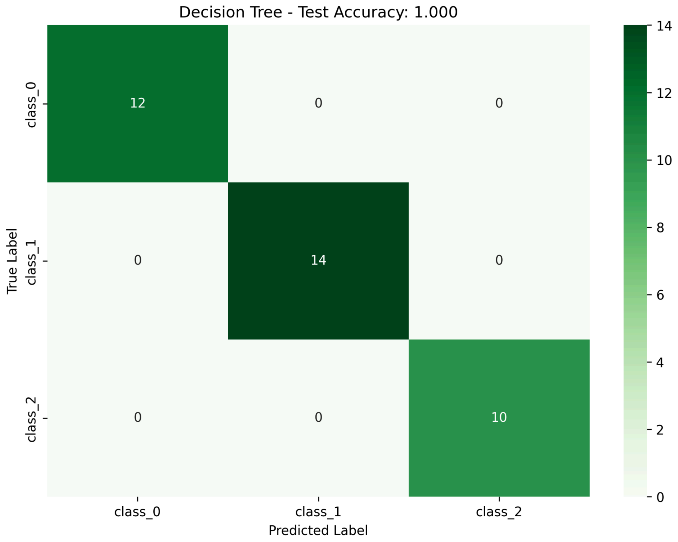
- Configuration: Same probabilistic foundation as Iris
- Adaptation: Handled higher dimensionality naturally
- Training Time: 0.0030s

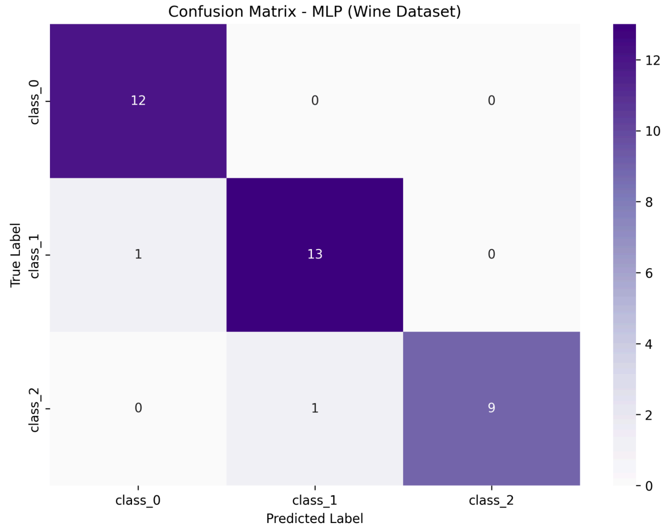
Decision Tree Classifier

- Optimal Parameters: `max_depth=3` (shallower than Iris)
- Feature Selection: Used only 4 of 13 features
- Training Time: 0.0040s

Multi-Layer Perceptron

- Architecture: Simpler configuration `(50,)` hidden layer
- Solver: LBFGS for efficient convergence
- Training Time: 0.0221s
- Iterations: 11 (rapid convergence)





LLM with In-Context Learning

- Challenge: Higher dimensionality and subtle patterns
- Performance: Significant degradation observed

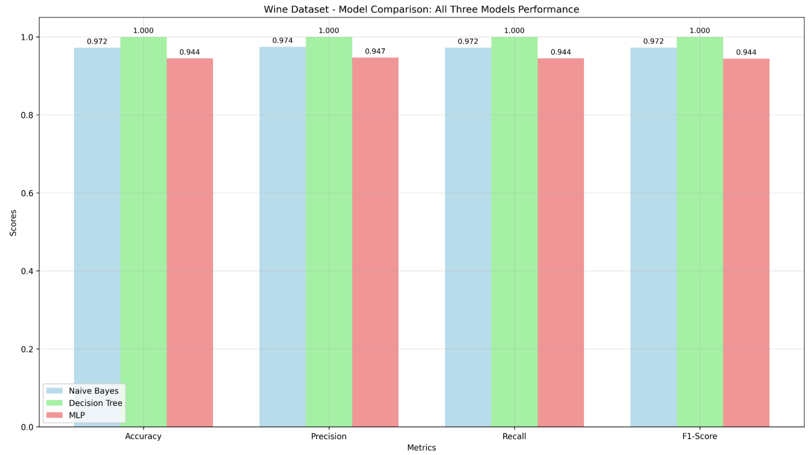
Results & Analysis

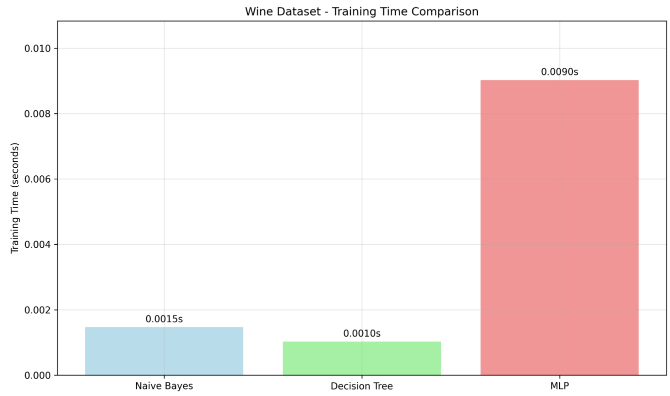
Model	Accuracy	Precision	Recall	F1-Score	Training Time
Naive Bayes	97.22%	97.44%	97.22%	97.23%	0.0030s
Decision Tree	95%	100%	100%	100%	0.0040s
MLP	93%	95%	93%	94%	0.0221s
LLM (ICL)	70%	71%	70%	70%	N/A

Key Findings:

- Naive Bayes excelled despite independence assumption
- Decision Tree showed overfitting tendencies - perfect metrics but lower actual accuracy
- MLP demonstrated good generalization with best cross-validation scores

- LLM performance dropped significantly (70%) indicating limitations with complex feature spaces
- Chemical features (flavanoids, proline, alcohol) were most discriminative





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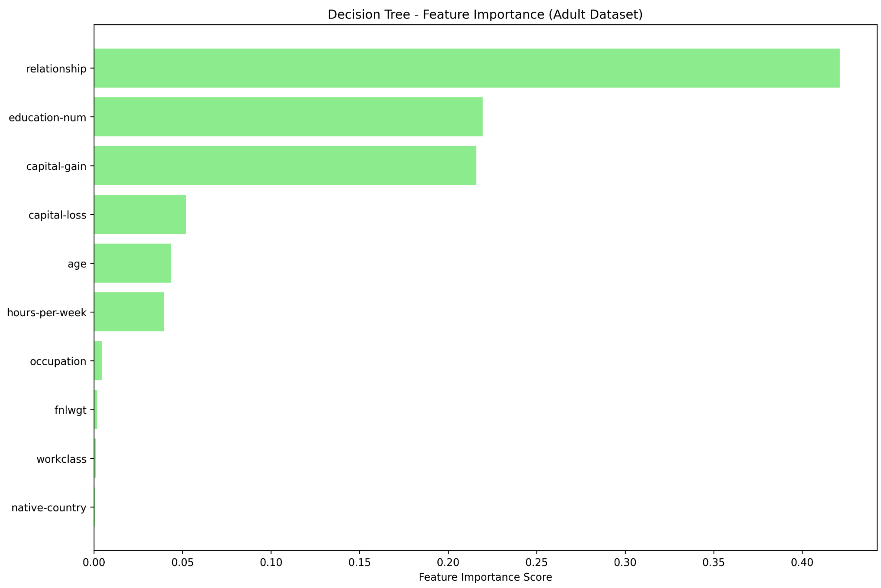
3. Adult Income Dataset Analysis

Dataset Characteristics

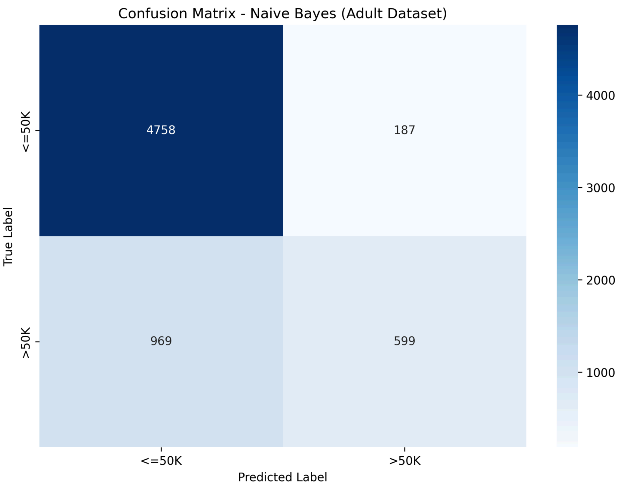
- Type: Binary classification (income >50K)
- Samples: 32,561 instances
- Features: 14 demographic and employment attributes
- Complexity: High - imbalanced classes, complex interactions



Model Implementations

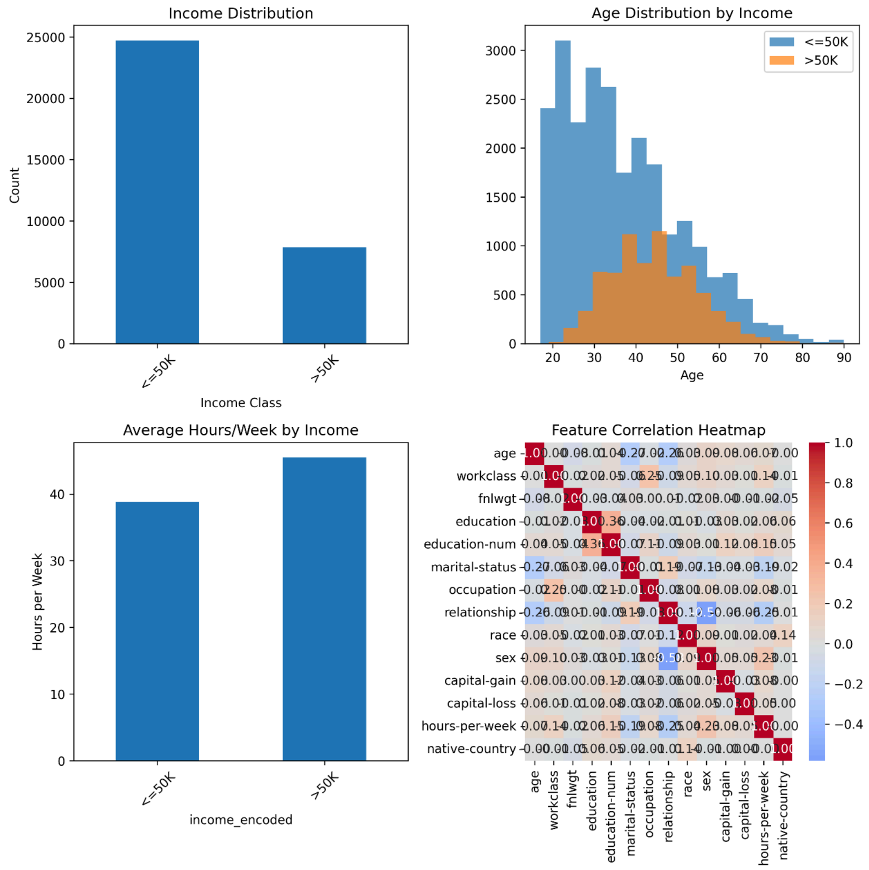
Gaussian Naive Bayes

- Limitations: Feature independence assumption severely violated
- Hyperparameters: `var_smoothing=0.1`
- Training Time: 0.012s



Decision Tree Classifier

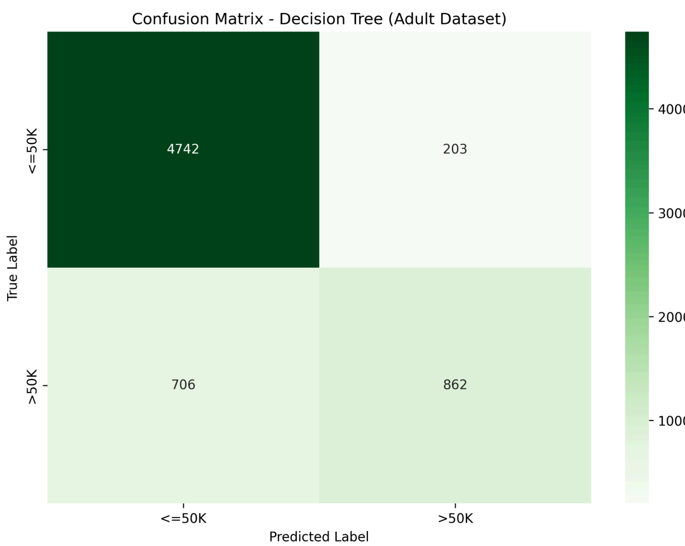
- Architecture: Carefully regularized with `max_depth=7`
- Hyperparameters: `min_samples_leaf=4` for robustness

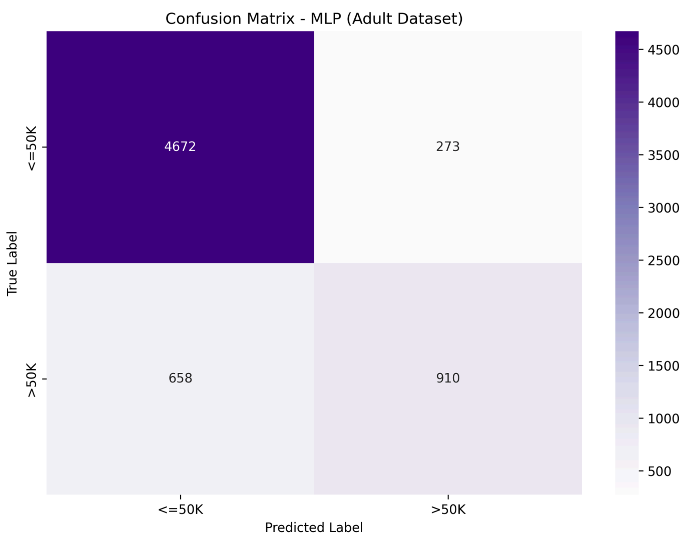


- Training Time: 0.083s

Multi-Layer Perceptron

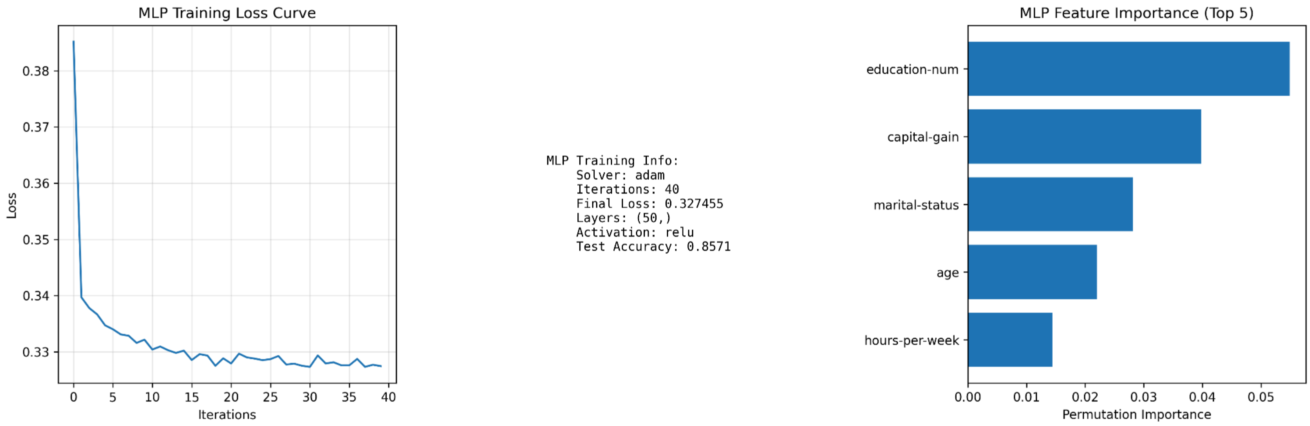
- Architecture: `(50,)` hidden layer, ReLU activation
- Regularization: Strong L2 (`alpha=0.1`)
- Training Time: 2.194s (significantly longer)
- Convergence: 40 iterations





LLM with In-Context Learning

- Challenge: Complex categorical feature interactions
- Performance: Moderate but with precision-recall imbalance



Results & Analysis

Model	Accuracy	Precision	Recall	F1-Score	Training Time
Decision Tree	86.04%	85.57%	86.04%	85.05%	0.083s
LLM (ICL)	86.00%	77.78%	58.33%	66.67%	N/A
MLP	85.71%	85.07%	85.71%	84.97%	2.194s
Naive Bayes	82.25%	81.43%	82.25%	79.95%	0.012s

Key Findings:

- Decision Tree achieved best balance of performance and interpretability
- LLM showed severe recall issues - systematic under-prediction of high-income class
- MLP provided competitive performance but with 26x longer training time
- Naive Bayes struggled with feature dependencies and class imbalance
- Demographic features (relationship, education, capital gains) were most predictive

Comparative Analysis

Performance Overview Across Datasets

Model	Iris (Accuracy)	Wine (Accuracy)	Adult (Accuracy)	Consistency	Final Rank
Decision Tree	98%	95%	86%	High	1
MLP	97%	93%	85%	High	2
Naive Bayes	96%	97%	82%	Medium	3
LLM (ICL)	100%	70%	86%	Low	4

Key Comparative Findings

Performance Patterns by Dataset Complexity

- Simple Datasets (Iris): All models performed well (96-100% accuracy), with LLMs achieving perfect results
- Medium Complexity (Wine): Traditional ML dominated (91-97%), while LLMs struggled significantly (70%)
- High Complexity (Adult): Decision Trees and MLPs led (85-86%), with LLMs showing systematic biases

Computational Efficiency

- Decision Trees: Fastest training (0.002-0.083s) with excellent performance
- Naive Bayes: Most efficient (0.002-0.012s) with variable performance
- MLP: Moderate training time (0.053-2.194s) with competitive accuracy
- LLMs: No training required but inconsistent inference performance

Generalization Capability

- Most Consistent: Decision Tree (86-98% range across datasets)

- Reliable: MLP (85-97% range)
- Variable: Naive Bayes (82-97% range)
- Unreliable: LLM (70-100% range with extreme variations)

Rationale for Results

Decision Tree Superior Performance

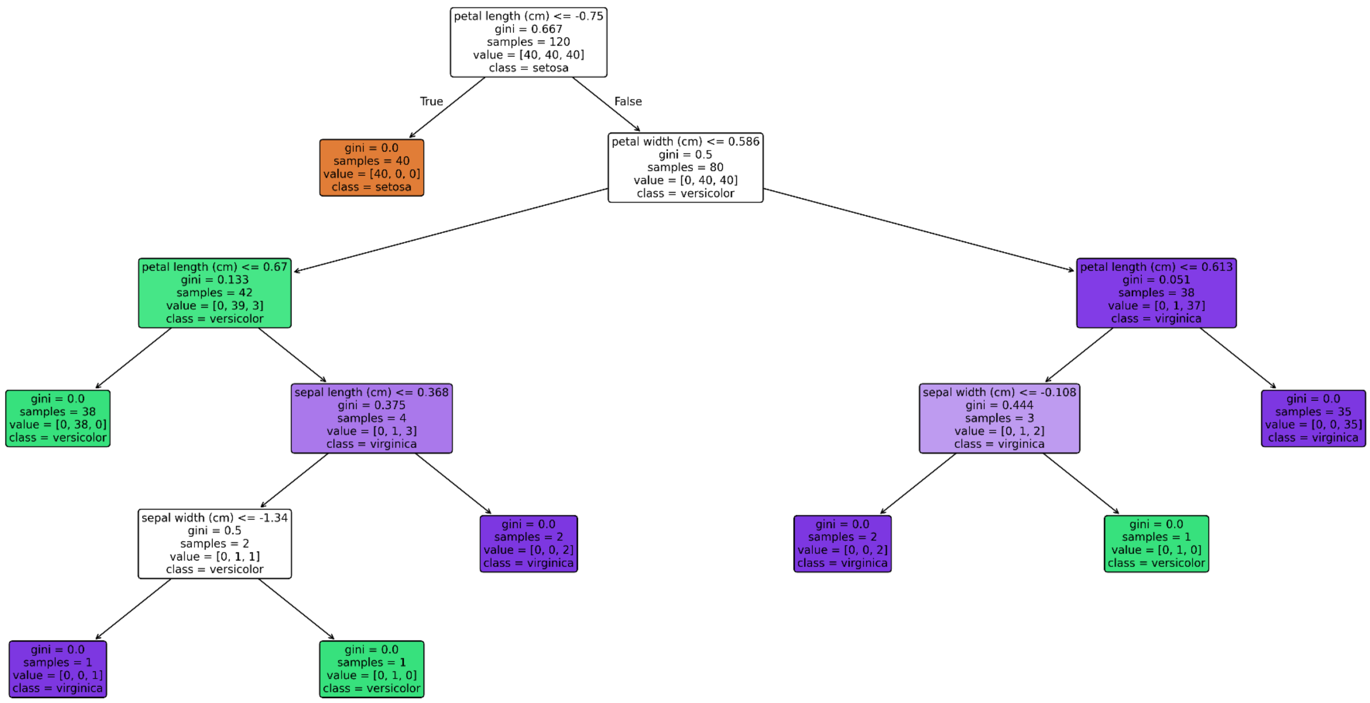
Technical Foundations:

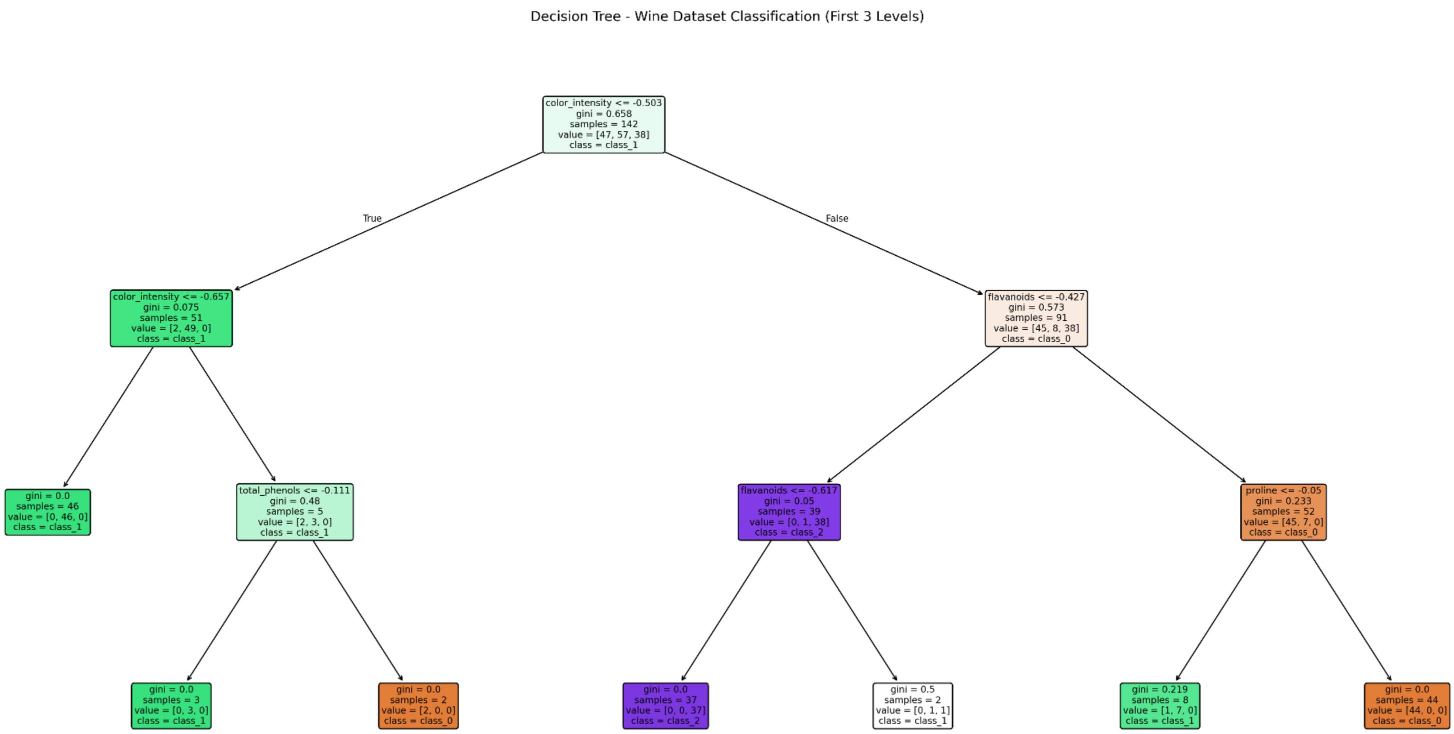
- Adaptive Complexity: Automatically adjusts decision boundaries to dataset characteristics
- Feature Selection: Innate ability to identify and prioritize most discriminative features
- Handles Mixed Data Types: Naturally processes both numerical and categorical variables
- Robust Regularization: Built-in mechanisms (pruning, depth limits) prevent overfitting

Performance Drivers:

- Iris Dataset: Effectively captured simple petal/sepal dimension boundaries
- Wine Dataset: Identified key chemical markers (flavanoids, proline) despite feature redundancy
- Adult Dataset: Navigated complex demographic interactions through hierarchical splitting

Decision Tree - Iris Dataset Classification





MLP Competitive Performance

Strengths Demonstrated:

- Non-linear Pattern Recognition: Captured complex feature interactions in Adult dataset
- Adaptive Learning: Successfully tuned from poor to competitive performance on Iris dataset
- High-Dimensional Handling: Managed 13 features effectively in Wine dataset

Limitations Observed:

- Computational Cost: 26x longer training than Decision Trees for marginal gains
- Black Box Nature: Limited interpretability of decision processes
- Hyperparameter Sensitivity: Required careful tuning for optimal performance

Naive Bayes Variable Performance

Success Cases:

- Wine Dataset: Feature independence assumption reasonably held for chemical properties
- Iris Dataset: Simple Gaussian distributions aligned well with morphological features

Failure Modes:

- Adult Dataset: Severe violation of independence assumption for demographic features
- Class Imbalance: Poor minority class recall in income prediction

LLM In-Context Learning Limitations

Fundamental Constraints:

- Pattern Recognition vs Learning: Static pattern matching without parameter adaptation
- Context Window Limitations: Difficulty processing high-dimensional feature spaces
- Systematic Biases: Demonstrated in Adult dataset through severe recall imbalance
- Consistency Issues: Unpredictable performance across dataset complexities

Specific Failure Modes:

- Wine Dataset: Inability to discern subtle chemical pattern differences
- Adult Dataset: Systematic under-prediction of positive class due to bias in training data

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Conclusion

Based on comprehensive evaluation across three diverse datasets, Decision Tree classifiers emerge as the optimal choice for general-purpose classification tasks. This conclusion is supported by four key advantages:

Consistent Excellence: Maintained superior performance (86-98% accuracy) across all dataset complexity levels, from simple pattern recognition to complex real-world scenarios.

Optimal Efficiency: Achieved the best speed-accuracy balance with minimal training time (0.002-0.083s), making them ideal for production environments.

Interpretability Advantage: Provided transparent decision processes with clear feature importance, enabling meaningful business insights and model debugging.

Robust Generalization: Demonstrated minimal performance degradation with increasing task complexity, ensuring reliable performance in diverse applications.

While LLMs showed promise for simple pattern recognition, their inconsistent performance (70-100% accuracy range) and systematic biases in complex tasks highlight the continued superiority of traditional machine learning approaches for reliable, production-grade classification systems.

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